10 PLATS AND MAPPING

MAP TYPES AND SYMBOLS

1. Which of the following statements does not correctly describe a difference between a map and a plat?

(A) A plat may function as a legal description of the land it represents in lieu of a written description. A map rarely fulfills such a function.

(B) A plat is more likely to show property boundaries than is a map. A map's emphasis is more often on physical features.

(C) A map may contain omissions or generalizations of particular physical features. A plat generally shows all data necessary to accurately represent the land it delineates.

(D) A map is more likely to show dimensional data than is a plat. A plat usually is intended for uses where distances and quantities are determined by scaling.

Solution: A map is much more likely to depend on scaling for the derivation of distances, quantities, etc., than is a plat. A plat should show, among other things, sufficient dimensional information to make scaling unnecessary.

Answer is (D)

2. Which of the following best defines the general characteristics of a planimetric map?

(A) a map showing the relief of an area by hachures or shading
(B) a map showing public areas and rights-of-way
(C) a map delineating different soils and geological features
(D) a map that shows only the relative horizontal positions of the features represented, without relief

Solution: A planimetric map is distinguished from a topographic map by the absence of any representation of the vertical orientation of the features shown.

Answer is (D)

3. On maps, a full-headed north arrow is sometimes accompanied by an arrow with half a head. What is the usual significance of such a half-headed arrow?

(A) It is used to indicate the direction of astronomic north.
(B) It is used to indicate the direction of magnetic north.
(C) It is used to indicate the direction of north when the orientation of the map makes it necessary for north to be shown other than toward the top of the sheet.
(D) It is used to indicate the direction of the basis of bearings.

Solution: The most frequent use of the half-headed arrow is to indicate magnetic north. Often, two arrows are shown; the angle between them is the declination.

Answer is (B)

4. Which of the following correctly describes an isogonic map?

(A) a map that indicates the areas within which particular zoning regulations apply
(B) a map that shows the relative positions of the boundaries of several other maps
(C) a map with lines connecting points of equal magnetic declination
(D) a map with a scale of 1:600,000 or smaller

Solution: The term *isogonic* refers to equal angles. An isogonic map includes lines of equal magnetic declination.

Answer is (C)

5. What type of map shows contours that connect points of equal rainfall?

(A) a hydrological map
(B) a bathymetric map
(C) an isohyetal map
(D) a Thiessen network

Solution: Contours are used in several types of maps besides topographic maps. Contours are used to indicate the mapping of gravity on maps that delineate the geoid and are used to connect points of equal rainfall on isohyetal maps. Maps that use lines joining points with equal numerical values are known collectively as *isoplethic maps*.

Answer is (C)

MAP SCALES

6. The legend of a map indicates that its scale is 1:24,000. What name is usually given to this type of expression of scale?

(A) an engineer's scale
(B) a graphical scale
(C) a manuscript fraction
(D) a representative fraction or scale ratio

Solution: The name *representative fraction* or *scale ratio* is usually given to this sort of dimensionless scale.

Answer is (D)

7. Which of the following scales is closest to 1 in = 1 mi?

(A) 1/62,500
(B) 1/24,000
(C) 1/100,250
(D) 1/63,360

Solution: Scale expressed as a representative fraction is independent of units of measurement. One unit on the map is intended to represent the stated number of the same units on the ground.

$$\begin{aligned} 1 \text{ mi} &= (5280 \text{ ft})\left(\frac{12 \text{ in}}{1 \text{ ft}}\right) \\ &= 63{,}360 \text{ in} \end{aligned}$$

The scale 1/62,500 is close to 1 in = 1 mi and is used as such on many maps.

Answer is (D)

8. A map's legend shows that it has a scale of 1:62,500. A measurement between two features on the map is read as 2.72 on a scale that is incremented at 20 parts/in. What distance does the measured length represent on the ground?

(A) 7083 ft
(B) 14,167 ft
(C) 170,000 ft
(D) 17,000 ft

Solution: The scale of 1:62,500 may be interpreted as 1 ft = 62,500 ft or as 1 in = 62,500 in.

$$\begin{aligned} 1 \text{ in} &= (62{,}500 \text{ in})\left(\frac{1 \text{ ft}}{12 \text{ in}}\right) \\ &= 5208.33 \text{ ft} \end{aligned}$$

The reading from a 20 parts/in scale can be divided by 2 to find the corresponding reading on a 10 parts/in scale.

$$\frac{2.72}{2} = 1.36$$

$$(5208.33 \text{ ft})(1.36) = 7083.33 \text{ ft}$$

Answer is (A)

9. What is the representative fraction or scale ratio for 1 cm = 500 m?

(A) 1:50,000
(B) 1:254
(C) 1:5000
(D) 1:24,000

Solution: One method of converting to representative fractions or scale ratios is to express both sides of the equation in the same units. There are 100 cm in a meter. Therefore,

$$500 \text{ m} = 50\,000 \text{ cm}$$

The equivalent representative fraction is 1:50,000.

Answer is (A)

10. A map prepared for an engineering project has a scale of 1 in = 100 ft. A map prepared for a land subdivision has a scale of 1 in = 1 mi. Which of the following statements is true?

(A) Both maps have a large scale.
(B) The engineering map has the smaller scale of the two.
(C) Both maps have a small scale.
(D) The engineering map has the larger scale of the two.

Solution: The more area a particular map represents, the smaller its scale. In other words, a large-scale map covers less area than a small-scale map.

Answer is (D)

TOPOGRAPHIC MAPPING

11. Which of the following best defines a topographic map?

(A) a map that shows only the relative horizontal positions of the features represented, without relief
(B) a map delineating different soils and geological features
(C) a map that shows the relative positions of the boundaries of several other maps
(D) a map showing both the horizontal and vertical locations of the features it represents

Solution: A topographic map shows the vertical as well as the horizontal locations of the features it represents. Generally, contours are used to indicate the relative elevations of the valleys, mountains, and plains shown.

Answer is (D)

12. What is a contour interval?

(A) a line of constant elevation above the reference ellipsoid
(B) the horizontal distance between adjacent contour lines
(C) one-half the vertical spacing of the contour lines specified in the legend of the map
(D) the difference in elevation between adjacent contours

Solution: A contour interval is the difference in elevation between adjacent contour lines. It is usually specified in the map's legend.

Answer is (D)

13. What is the primary use of interpolation in constructing a topographic map?

(A) the connection of spot elevations by contour lines
(B) the determination of intermediate contour lines between points or lines of known elevation
(C) the designation of index contours
(D) drawing hachures

Solution: Interpolation is the determination of intermediate values between two given values. Where a constant rate of change of the slope can be assumed between points of known elevation, the process is used to locate intermediate contour lines.

Answer is (B)

14. In an area of a topographic map where the contour lines are equally spaced, a straight line crosses several contour lines. A consistent distance of 0.22 in is scaled on the map between each intersection of the straight line and the successive contour lines. If the scale of the map is 1:24,000 and the contour interval is 20 ft, what is the slope of the line?

(A) 4.55%
(B) 0.38%
(C) 2.47%
(D) 3:1

Solution: The horizontal distance on the ground represented by 0.22 in on the map can be found as follows. According to the map's scale,

$$\begin{aligned} 1 \text{ in} &= 24{,}000 \text{ in} \\ &= 2000 \text{ ft} \\ (2000 \text{ ft})(0.22) &= 440 \text{ ft} \end{aligned}$$

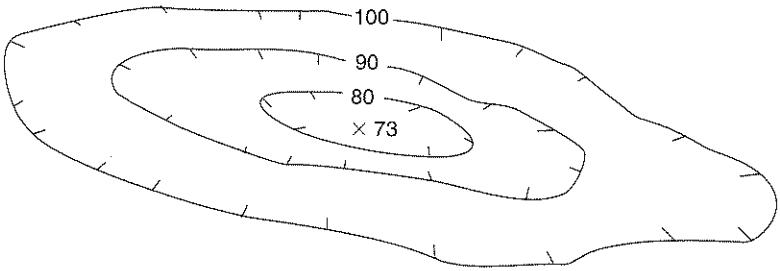
The horizontal distance on the ground between successive intersections of the straight line and the contours it crosses is 440 ft.

The contour interval on the map is 20 ft. The division of the contour interval by the distance of the line between the contours, 440 ft, yields the slope of the line.

$$\frac{20 \text{ ft}}{440 \text{ ft}} = 0.0455 = 4.55\%$$

Answer is (A)

15. What sort of topographic feature is indicated by the illustration shown?



(A) a lake or pond
(B) a tailings pile
(C) a glacier
(D) a depression

Solution: Closed contours with tick marks or hachures inside the figure are used to indicate a depression. It is best to show the elevation of the lowest point of such a figure as well.

Answer is (D)

16. Which of the following statements is true of contour lines on a topographic map?

(A) Under some circumstances, a single contour line may split into two.
(B) Under some circumstances, two contour lines may join into one.
(C) A contour line cannot end within a map; it must either close on itself or terminate at the neat line of the map.
(D) One contour line never crosses another.

Solution: Contour lines cannot join, nor can one contour line split into two. However, it is sometimes possible for them to cross, as in the case of overhanging ledges, where the lower contours may be shown as dashed lines. Contour lines either close on themselves or end at the edge of the map; they cannot stop within the boundaries of the map.

Answer is (C)

17. The preparation of a topographic map requires that 5-ft contours be drawn. Two points, A and B, are 3.67 in apart on the map. Point A has an elevation of 773.34 ft, and point B has an elevation of 801.67 ft. On the map, what is the distance along the line between A and B from point A to the 790.00-ft contour?

(A) 2.72 in
(B) 1.66 in
(C) 2.16 in
(D) 1.91 in

Solution: The vertical distance between points A and B is 28.33 ft, and the horizontal distance between the same two points on the map is 3.67 in. Assuming that the grade between the two points is relatively constant, the rate of change in elevation between the two points is

$$\frac{3.67 \text{ in}}{28.33 \text{ ft}}$$

The vertical distance from point A to the 790.00-ft contour is 16.66 ft. The horizontal distance, x, on the map can then be found by proportion.

$$\begin{aligned}\frac{3.67 \text{ in}}{28.33 \text{ ft}} &= \frac{x}{16.66 \text{ ft}} \\ x &= \frac{(16.66 \text{ ft})(3.67 \text{ in})}{28.33 \text{ ft}} \\ &= 2.16 \text{ in}\end{aligned}$$

Answer is (C)

PLATS

18. Some general features of the laws governing platting are the same from state to state, but the details vary considerably. Which of the following specifications is most likely to differ from jurisdiction to jurisdiction?

(A) The boundary of the tract must be clearly shown with permanent monuments marking its corners.
(B) The surveyor's certification must include an official seal and signature.
(C) Enough information regarding the lengths and directions of the lines must be provided so that another surveyor can retrace them correctly.
(D) A surveyor must prepare a plat for every survey performed.

Solution: While some states specify that a plat be recorded for every survey performed, many do not.

Answer is (D)

19. Which of the following describes objectives that should be satisfied by a properly prepared plat?

(A) a representation of the correct shape and size of the property surveyed at scale
(B) the description and relative location of pertinent monuments, whether physical or record
(C) the identification of the appropriate evidence of title
(D) all of the above

Solution: While a plat may not settle questions of title, it should provide evidence of easements and encroachments, as well as other record and physical information affecting the title of the property.

Answer is (D)

20. Which type of scale on a plat is most useful, and why?

(A) A representative fraction is most useful because it is dimensionless.

(B) A ratio scale, such as 1 in = 500 ft, is most useful since such a scale is universally understood.

(C) A bar scale is most useful since it remains valid despite enlargement or reduction of the plat.

(D) An architect's scale, such as $\frac{1}{8}$ in = 1 ft, is most useful since it allows the representation of the greatest detail.

Solution: The bar scale, required by many platting laws, has the unique characteristic of retaining its validity despite reduction or enlargement of the plat.

Answer is (C)

21. When do the monuments shown on a plat have equal standing with the monuments mentioned in a deed?

(A) when the plat is a matter of public record

(B) when the plat was drawn prior to the writing of the deed and the deed is unrecorded

(C) when the deed refers to the plat and the parties to the deed acted with reference to the plat

(D) when the monuments shown on the plat were set after the deed was written

Solution: When parties to the deed act in reference to the plat, it gives status to the monuments shown on the plat. When the plat is mentioned in the writings of the deed, the plat and the monuments shown on it become an integral part of the deed itself.

Answer is (C)

22. When a plat is filed by the county recorder, which of the following automatically ensues?

(A) Constructive notice is implied.
(B) The title insurer's liability ends.
(C) The bearings and distances shown supersede the previous measurements.
(D) The title is transferred.

Solution: Recording a plat in the public registry automatically implies constructive notice. The plat becomes available to all who wish to know it.

Answer is (A)

ALTA/ACSM LAND TITLE SURVEY PLATS

23. Which of the following is not a minimum standard detail requirement for the plat of an ALTA/ACSM Land Title Survey?

(A) North will be toward the top of the plat.
(B) A graphic scale will be included.
(C) The minimum size will be 18 in $\times$ 26 in with a 1-in border.
(D) The radius, chord, and bearing of each horizontal curve will be shown.

Solution: The size described in the minimum standard detail requirements for ALTA/ACSM Land Title Survey plats is $8\frac{1}{2}$ in $\times$ 11 in.

Answer is (C)

24. Which statement correctly describes the minimum requirement regardin the delivery of the plat of an ALTA/ACSM Land Title Survey to the client titl insurance company?

(A) The surveyor must furnish three copies of the plat and the associate boundary description, which must be attached to the face of the plat i feasible.

(B) The surveyor must furnish two copies of the plat drawn in ink on myla and four copies of the associated boundary description on separate sheets

(C) The surveyor must furnish two copies of the plat and the associate boundary description, which must be attached to the face of the pla if feasible. If multiple sheets are required, each sheet will be numbered the total number of sheets will be indicated on each, and match lines wil be shown.

(D) The surveyor must furnish two copies of the plat drawn in ink on myla and four copies of the associated boundary description on separate sheets The plat must be under the title *ALTA/ACSM Land Survey Title Plat* and the section, township, and range must be indicated on the plat.

Solution: Only some of the requirements for delivery of the plat are listed here Two copies of the plat and description are required. The description shoul be attached to the plat if feasible. The pages of the plat should be numbere consecutively and the total number of pages indicated on each. The requirement do not specify the medium on which the plat must be drawn, except to say tha it must be stable and acceptable to the title insurance company.

Answer is (C)

25. The location of all buildings on the parcel must be shown on the plat of an ALTA/ACSM Land Title Survey. Which of the following statements correctly describes the measurements required to locate the buildings?

(A) The measured ties locating the buildings will be shown perpendicular to the boundaries.

(B) The measured ties locating the buildings will be shown directly to the nearest verified property corners.

(C) The measured ties locating the buildings will be shown along the lines established by the building's walls.

(D) The measured ties locating the buildings will be shown perpendicular to the building's wall lines.

Solution: Ties to buildings on the parcel are to be shown perpendicular to the property boundaries.

Answer is (A)

26. The location and type of walls, fences, and buildings must be shown on the plat of an ALTA/ACSM Land Title Survey within what distance on either side of the boundary lines?

(A) 1 ft
(B) 2 ft
(C) 3 ft
(D) 4 ft

Solution: All possible encroachments from adjoining properties must be shown. In addition, all fences, walls, and buildings within 2 ft on either side of the property boundary are to be shown.

Answer is (B)

27. A surveyor may note the date of the measurements taken and that a boundary shown on the plat of an ALTA/ACSM Land Title Survey is subject to change under which of the following circumstances?

(A) The boundary is apparently encumbered by an easement.
(B) The boundary abuts a railroad right-of-way.
(C) A natural water boundary is involved.
(D) The monuments found do not agree with the record.

Solution: The location of property dependent on a water boundary is subject to change by natural causes and should be so noted on the plat of an ALTA/ACSM Land Title Survey.

Answer is (C)

28. The specifications for the degree of precision required of an ALTA/ACSM Land Title Survey are based on what principle?

(A) The degree of precision should be based on the current use and development of the property.
(B) The degree of precision should be based on the availability of control in the vicinity of the property.
(C) The degree of precision should be based on the extent of the surveyor's professional liability.
(D) The degree of precision should be based on the intended future use of the property.

Solution: The surveyor should consider the likely use of the property in the future, without regard to its present state, when gauging the precision required for a particular survey.

Answer is (D)

29. Which of the following is stipulated in the ALTA/ACSM Minimum Standard Detail Requirements for surveys of residential subdivisions in suburban areas?

(A) direct reading of $10''$ if an electronic theodolite is used
two direct and reverse positions observed for the angles at each station
the angular closure not to exceed $10''\sqrt{N}$ (N = the number of stations)
the relative precision of distance measurements will not exceed 1:15,000

(B) direct reading of $10''$ if an electronic theodolite is used
two direct and reverse positions observed for the angles at each station
the angular closure not to exceed $15''\sqrt{N}$ (N = the number of stations)
the relative precision of distance measurements will not exceed 1:10,000

(C) direct reading of $20''$ if an electronic theodolite is used
one direct and reverse position observed for the angles at each station
the angular closure not to exceed $20''\sqrt{N}$ (N = the number of stations)
the relative precision of distance measurements will not exceed 1:7500

(D) direct reading of $1'$ if an electronic theodolite is used
one direct and reverse position observed for the angles at each station
the angular closure not to exceed $30''\sqrt{N}$ (N = the number of stations)
the relative precision of distance measurements will not exceed 1:5000

Solution: The specifications listed are partial for the four classes of surveys outlined in the ALTA/ACSM Minimum Standard Detail Requirements for surveys. The standards are labeled as they are in the ALTA/ACSM specifications—that is, A, B, C, and D. The B standard is appropriate to suburban surveys.

Answer is (B)